LAB EXPERIMENT - 9

EXPERIMENT 9: MITRE ATT&CK CLASSIFICATION

➢ Aim

To map cyber threats to the MITRE ATT&CK framework.

Inputs

- Indicators of compromise (IoCs)
- Attack behavior

Tools Used

- MITRE ATT&CK Navigator (simulated)

Dataset Required

- Threat data from SolarWinds attack

➢ Brief Theory

MITRE ATT&CK is a framework used to classify cyber attacker behavior based on tactics and techniques. It helps in understanding how attacks are carried out and improves detection strategies.

➢ Procedure

1. Identified attack behaviors from SolarWinds incident
2. Mapped behaviors to MITRE ATT&CK tactics
3. Classified techniques under each tactic
4. Created ATT&CK mapping matrix

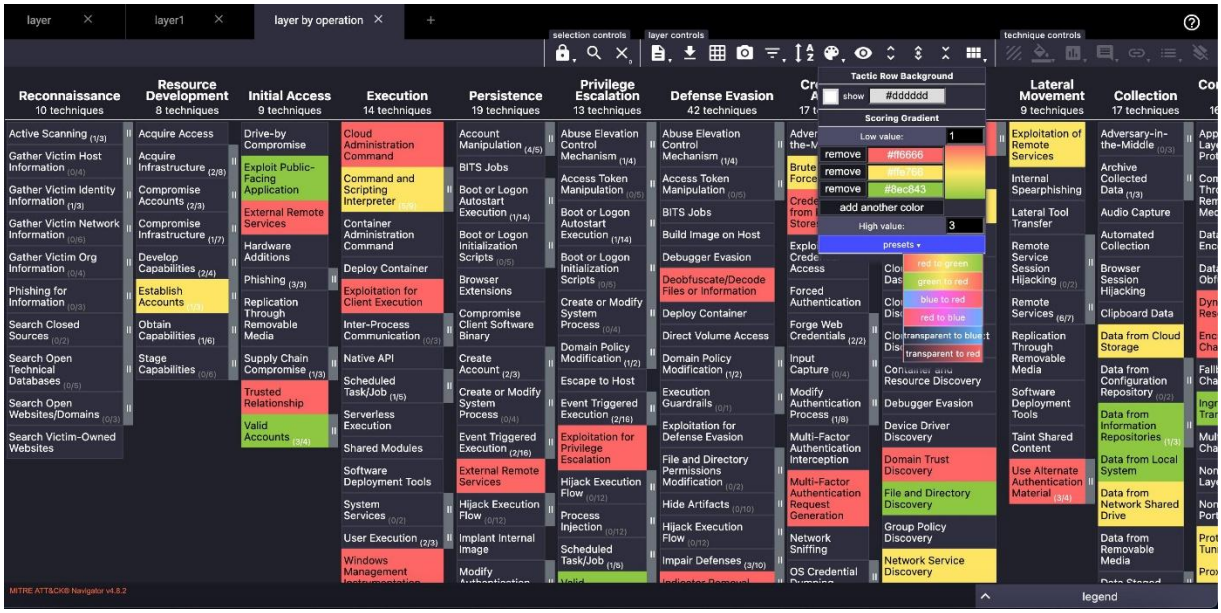
- **Attack Selected**

SolarWinds Supply Chain Attack

➢ MITRE ATT&CK Mapping Table

Tactic	Technique	Description
Initial Access	Supply Chain Compromise	Malicious code inserted into software updates
Execution	Command Execution	Malware executed after installation
Persistence	Backdoor Access	Attackers maintained long-term access
Privilege Escalation	Credential Theft	Stole admin credentials
Lateral Movement	Remote Access	Spread across systems
Collection	Data Collection	Gathered sensitive data
Exfiltration	Data Exfiltration	Sent data to attacker servers

➢ MITRE ATT&CK Heatmap Visualization



➢ Analysis

- The attack involved multiple stages across the ATT&CK framework
- Supply chain compromise was the initial entry point
- Attackers used stealth techniques for persistence and data theft

➢ Result

The SolarWinds attack was successfully mapped to MITRE ATT&CK tactics and techniques.

➢ Conclusion

MITRE ATT&CK helps in understanding attacker behavior and improves threat detection strategies.